Project WalkThrough

CodeBase

ML Basics

Report

1. The Core Problem (Introduction)

Amyotrophic Lateral Sclerosis (ALS) is a severe disease that breaks down motor neurons in the nervous system. Right now, doctors struggle to diagnose it early because there is no simple, non-invasive blood test.

Scientists can look at blood using "transcriptomics" (studying which genes are turned on or off), but the genetic signals for ALS in the blood are incredibly weak and scattered.

Because the signals are so quiet, a single study or a single dataset usually doesn't have enough statistical power to find them reliably. To fix this, your project aimed to combine massive amounts of data from different sources to create one powerful, universal "molecular fingerprint" for ALS.

2. The Data (Materials)

The project used two different datasets of peripheral whole blood, representing two different generations of genetic technology:

- Microarray (Older Tech): Think of this like taking a photograph of the genes using chemical tags that glow. The dataset (GSE112681) had 1,115 samples.
- RNA-Sequencing (Newer Tech): Think of this like literally counting every single genetic letter. The dataset (GSE234297) had 144 samples.

3. The Pipeline (Pre-processing & Harmonization)

Combining these two datasets is like trying to compare a digital photograph with a hand-drawn painting; they measure the same thing but look completely different mathematically.

Here is how the project solved that:

- Finding Common Ground: The team matched the genes between the two datasets, finding an overlapping space of 9,344 shared genes.
- Batch-Effect Correction (ComBat): This is the most crucial step. When you merge data from two different machines, the mathematical differences caused by the machines (technical noise) can drown out the actual biological differences (the disease). The team used an algorithm called ComBat to digitally erase the machine-specific biases while preserving the true ALS signals.

4. Biological Analysis (Finding the Genes)

Once the data was combined and cleaned, the team used statistical math (a tool called limma) to find which genes were behaving differently in ALS patients versus healthy people.

- The Fingerprint: They found a strict "core" of 4 highly significant genes, a medium-confidence group of about 90 genes, and a broader set of 500 genes to use for machine learning.
- Network Analysis (WGCNA): The team tried to see if these genes grouped together into functional "neighborhoods" or modules. However, because the ALS signal in blood is so subtle, this network analysis only found one weak group and wasn't very useful.

5. Machine Learning (The Ultimate Test)

To prove this 500-gene fingerprint actually works, the team set up a brutal, cross-platform test. They trained Artificial Intelligence models using *only* the RNA-seq data, and then tested them on the completely unseen Microarray data. This ensures the AI is learning the actual disease, not just memorizing the data.

They tested three AI models:

1. Random Forest: A complex "council" of decision trees. It scored an AUC of 0.669 (AUC is a grade from 0.5 to 1.0, where 1.0 is perfect). It struggled slightly because it over-complicated things and memorized some technical noise.
2. Support Vector Machine (SVM): A model that tries to draw a mathematical boundary between healthy and sick patients. It did slightly better with an AUC of 0.682.
3. Logistic Regression: A simpler, linear mathematical scale. Surprisingly, this simple model performed the best, scoring an AUC of 0.712!. It proved that for subtle genetic signals, a simple, straightforward model is less likely to get confused by noise.

6. The Ablation Study (Finding the Sweet Spot)

Finally, the team did an "Ablation Study" on the winning Logistic Regression model. This means they slowly stripped away genes to see how many were *actually* needed.

They found a "Goldilocks zone": exactly 25 genes provided the highest accuracy (an AUC of 0.746). Using 50 or 100 genes actually made the model worse because it started picking up unnecessary technical noise.

Summary

In short, this project successfully proved that by carefully cleaning and combining different types of blood data, we can use a simple machine learning model to find a reliable 25-gene signature that spots ALS, paving the way for easier diagnoses.

Abstract

Amyotrophic Lateral Sclerosis (ALS) is hard to diagnose early because the disease's genetic signs in the blood are extremely faint and scattered. To fix this, the project mathematically combined two massive, different sets of genetic data to filter out technical errors and make those hidden biological signals louder. By training machine learning (AI) on this cleaned-up data, the study successfully found a "molecular fingerprint" of genes that can help identify the disease. This proves that combining different types of data can pave the way for a simple, non-invasive blood test to diagnose ALS

Introduction

Amyotrophic lateral sclerosis (ALS) is a devastating nervous system disease, and early clinical intervention is hindered by the lack of reliable, non-invasive blood biomarkers. While high-throughput transcriptomics offers a way to profile molecules, the biological indicators of ALS in peripheral blood are extremely faint and widely distributed. Consequently, single-platform studies—using only microarrays or only RNA-sequencing—lack the statistical power to reliably detect these elusive signals. Although merging data from multiple platforms can improve detection, this process inherently introduces a dramatic amount of non-biological technical variation. Merging datasets without rigorous algorithmic batch-effect correction drastically inflates false-positive rates and obscures true biological signals. To overcome these challenges, this study introduces a multidimensional computational pipeline that seamlessly integrates and batch-corrects large-scale RNA-seq and microarray transcriptomes. This rigorous algorithmic approach resolves both the sensitivity limits of single-cohort studies and the technical hurdles of uncorrected data merging. Ultimately, this methodology yields a highly reliable, all-encompassing molecular signature that can be used universally for the non-invasive diagnosis of ALS.

Multi-Omics Data Integration: : Micro Array —> blood samples tested with old technology

RNA sequencing ——--> blood samples tested with newer technology

Cross-Platform Harmonization : Because the two testing technologies label things differently, the team had to create a standardized map. This allowed them to find an overlapping "common ground" of 9,344 genes that both datasets shared.

Erasing the Machine Noise (Algorithmic Batch Correction):

: **Differential Gene Expression (DGE)** is the statistical and biological process of identifying which specific genes are "turned up" (upregulated) or "turned down" (downregulated) when comparing two different conditions—such as a healthy control group versus a disease state.

Only exactly **4 genes** showed a mathematically strong, undeniable difference between healthy people and ALS patients

The **4 genes** prove the disease is there. The **90 genes** try to show how the disease operates. The **500 genes** give the AI enough puzzle pieces to actually diagnose the patient.

Materials & Method

The Microarray Dataset: **1115**

The RNA-Seq Dataset: This was your smaller, newer dataset containing 144 blood samples

In both datasets, the ultimate goal was to classify each sample as either "ALS" (sick) or "CON" (healthy control)

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Gene Intersection: The pipeline found the overlapping "common ground" between the two sets, resulting in 9,344 shared genes

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Differential Gene Expression (DGE) using "limma"

Because the biological signals for ALS in the blood are extremely faint, `limma` didn't just give you one simple list of "ALS genes".

When you ran the WGCNA tool on the ALS genes, the algorithm couldn't find any strong, obvious "cliques".

Dataset Description

You specifically chose peripheral whole blood because the ultimate goal is to create a non-invasive diagnostic tool

The Features (Predictor Variables - X): This is what the AI is actually looking at. In the older microarray data, these are continuous "fluorescence intensity" values (how brightly the gene glowed). In the newer RNA-seq data, these start as discrete "read counts" (how many times the gene was seen) which you mathematically transformed into a logarithmic scale (log_2) so both datasets could be compared

The Target (Variable - y): This is the ultimate answer the AI needs to guess: is the sample strictly an "ALS" patient or a healthy Control ("CON")?.

Why as RF used

: It is great at preventing "overfitting" (memorizing the data instead of learning from it) when you have more genes than actual patients. It also calculates an intrinsic "feature importance score," which tells you exactly which genes matter most.

Hyperparameters (The Settings): * `n_estimators = 1000`: You built 1,000 independent decision trees to ensure the results were statistically stable.

- `max_depth = 10`: You restricted how deep the trees could think. This acts as a "regularization mechanism" to stop the AI from memorizing the technical noise in the smaller RNA-seq dataset.
- `class_weight = 'balanced'`: Because you had more ALS cases than healthy controls in the training set, this setting forced the AI to pay equal attention to the minority group.

Why was Logistic Regression used

Despite being simple, it is often the most accurate tool in biology when the disease signal is incredibly subtle and spread across multiple genes like an additive equation.

Why was SVM used

It is incredibly effective for datasets where the difference between classes isn't a simple straight line

Performance Evaluation

Train korsi : 144 tar upor

Test korsi :1115 er upor

False Discovery Rate (FDR - Benjamini-Hochberg) to ensure your false positive rate stayed below 5%

Gini Impurity (Feature Importance): This is used as the main splitting criterion for the Random Forest classifier during training, as an error function to be minimised by the decision trees.

You could not use standard "accuracy" percentages to grade the AI because your test dataset was highly imbalanced (396 ALS vs. 719 Controls). Instead, you used the Area Under the Receiver Operating Characteristic Curve (AUC-ROC). This metric evaluates the model's true discriminatory power regardless of the class imbalance, making it a much fairer grade

ROC Curve :

Experimental Analysis

Q/A

Section 1: Core Concepts & The Problem Statement

1. What is the primary clinical limitation in ALS diagnosis that your project aims to address?

Answer: The lack of reliable, non-invasive, blood-based diagnostic biomarkers, which delays immediate intervention.

2. Why does analyzing a single transcriptomic platform (like only microarrays) fail to reliably detect ALS signals in the blood?

Answer: Because biological signals in peripheral blood are extremely subtle and widely distributed; therefore, single-modality studies often do not have enough statistical power to reliably detect these elusive systemic signals.

3. How does your multidimensional computational pipeline propose to overcome the statistical hurdles of single-cohort studies?

Answer: By seamlessly integrating large-scale peripheral blood microarray and RNA-seq datasets using rigorous variance stabilization, cross-platform gene symbol harmonization, and algorithmic batch correction.

4. What happens when high-throughput transcriptomic datasets are merged naively without algorithmic batch correction?

Answer: Naïve data merging drastically inflates false-positive rates and confounds true biological signals.

5. Briefly describe the three distinct tiers of the multi-level molecular fingerprint your pipeline delineated.

Answer: It includes a high-confidence core set of 4 genes, a medium-confidence functional fingerprint of roughly 90 genes, and a broader 500-feature predictive signature used for machine learning.

Section 2: Datasets & Features

6. Which two specific dataset accessions did you utilize, and what underlying technologies do they represent?

Answer: GSE112681, which represents microarray technology (Illumina HumanHT-12 V3.0 and V4.0), and GSE234297, which represents high-throughput RNA-sequencing technology.

7. Why was peripheral whole blood specifically chosen as the tissue source for this project?

Answer: It is the desired substrate for developing a non-invasive diagnostic tool.

8. Can you break down the sample sizes for ALS patients versus Healthy Controls in the Microarray (GSE112681) dataset?

Answer: Out of 1,115 total samples, there were 396 ALS patients and 719 Controls/MIMs.

9. What was the sample breakdown for the RNA-seq (GSE234297) dataset?

Answer: Out of 144 total samples, there were 96 ALS patients and 48 Controls.

10. Before cross-platform harmonization, how many raw genes/probes were measured in each dataset?

Answer: The microarray dataset measured 28,830 probes (mapped to 13,582 genes), and the RNA-seq dataset measured 46,025 raw features (mapped to 12,969 genes).

11. How many shared genes were retained to form your final harmonized expression matrix?

Answer: 9,344 shared genes.

12. In your Microarray data, what exactly do the predictor variables (X) represent physically?

Answer: They represent continuous fluorescence intensity values derived from hybridized probes.

13. How do the raw predictor variables in the RNA-seq data differ from the Microarray data?

Answer: They initially represent discrete read counts mapping to Entrez IDs, rather than continuous fluorescence values.

14. What specific target variable (y) were your models trying to predict?

Answer: The binary diagnosis status, categorized strictly as either "ALS" (sick) or "CON" (healthy controls and neurological mimics).

15. What does the "long right tail" in the density profile of the V4 Microarray data indicate biologically?

Answer: It is indicative of highly expressed housekeeping genes.

Section 3: Data Preprocessing & Harmonization

16. Walk me through the specific preprocessing steps applied to the Microarray cohort before merging.

Answer: The steps included rigorous quality control (QC) and outlier removal, followed by log_2 and quantile normalization, probe filtering, and intra-platform batch correction to merge V3 and V4 .

17. How did you mathematically handle the discrete read counts in the RNA-seq data so they could be compared to continuous microarray data?

Answer: They were transformed into continuous log_2-scaled values.

18. When mapping gene symbols across platforms, how did you handle duplicate features in the RNA-seq data?

Answer: Duplicates were handled using the median.

19. What specific gene nomenclature system was used to map the RNA-seq features?

Answer: Features initially mapping to Entrez IDs were mapped to matching official HUGO symbols to align with the microarray data.

20. Explain the core function of the ComBat algorithm in your pipeline.

Answer: ComBat was globally applied to systematically model and remove severe platform-specific technical variations (batch effects) while strictly preserving the true biological variance between ALS patients and healthy controls.

21. What would have happened to your biological variance (the ALS signal) if you hadn't used ComBat during integration?

Answer: Uncorrected data merging would drastically inflate false-positive rates and confound true biological signals.

22. Based on your data distribution analysis, what was the mean expression value of the RNA-seq dataset post-transformation?

Answer: The mean was 6.155.

Section 4: Biological Analysis (DGE & WGCNA)

23. Which specific statistical pipeline was used to conduct the Differential Gene Expression (DGE) analysis?

Answer: The limma pipeline, utilizing rigorous linear modeling.

24. How many genes in your results exhibited a strong effect size ($|log_2FC| > 0.5$)?

Answer: Only 4 genes exhibited a strong effect size.

25. What statistical metric and threshold did you use to control for false positives when testing over 9,000 genes simultaneously?

Answer: The False Discovery Rate (FDR - Benjamini-Hochberg) was used, ensuring the percentage of false positives was no greater than 5% ($FDR < 0.05$).

26. What does the incredibly small number of genes with a strong effect size tell us about the nature of ALS biomarkers in blood?

Answer: It confirms that ALS transcriptomic alterations in peripheral blood are inherently weak, subtle, and highly distributed.

27. Why did you apply Weighted Gene Co-expression Network Analysis (WGCNA) to your 90-gene fingerprint?

Answer: To evaluate the strength and limitations of co-expression modules within highly heterogeneous, multi-cohort data.

28. What soft-thresholding power was ultimately selected for the WGCNA model, and why?

Answer: Power 6 was selected to maintain a balance between scale-free topology connectivity and modular emergence.

29. Did the network exhibit strong scale-free properties? How do you know?

Answer: No, the scale-free topology fit (R^2) failed to reach the standard threshold of 0.80 at any power tested.

30. Why was the WGCNA investigation not pursued further in your pipeline?

Answer: Because the distributed cross-platform signal was too weak for standard modular clustering, yielding only a single weak topological block.

Section 5: Machine Learning Architecture

31. Why did you deliberately choose not to use a standard 80/20 train-test split on a single dataset?

Answer: Standard splits were avoided to test the true cross-platform generalizability of the molecular fingerprint.

32. Explain your platform-based partitioning strategy for the machine learning models.

Answer: The models were trained exclusively on the harmonized RNA-sequencing cohort and tested on a completely independent, unseen test set: the Microarray cohort.

33. Why was the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) chosen over standard accuracy for evaluation? Answer: Standard accuracy is inappropriate due to the high class imbalance of the Microarray test set (396 ALS vs. 719 Controls); AUC assesses discriminatory ability regardless of class distribution.

34. Why was Random Forest selected as your primary ensemble learning model?

Answer: It has high anti-overfitting ability in high-dimensional feature spaces, works well with non-linear biological relationships, and generates an intrinsic feature importance score.

35. How did the max_depth = 10 parameter in your Random Forest act as a regularization mechanism?

Answer: Restricting tree depth prevented the model from memorizing technical noise in the small RNA-seq training set, ensuring generalizability to the larger microarray test set.

36. Your training cohort was highly imbalanced. How did you mathematically adjust for this in your RF and LR models?

Answer: The parameter class_weight = 'balanced' was used to adjust classification penalties inversely proportional to class frequencies, ensuring the model did not ignore the minority group.

37. Why was Logistic Regression included as a baseline, and what was its max_iter parameter set to? Answer: It was selected for its high-performance linear baseline, as it often produces more accurate gene-level predictions when the biological signal is spread across multiple genes in an additive fashion. max_iter was set to 1000 to ensure full convergence.

38. What is the fundamental mathematical goal of the Support Vector Machine (SVM) model in your high-dimensional space?

Answer: To identify the hyperplane that maximizes the distance (margin) between the ALS and Healthy Control samples.

39. Why was the probability = True hyperparameter necessary for the SVM?

Answer: It was enabled to allow the calculation of class-membership probabilities via Platt scaling, which was necessary to generate ROC curves and calculate the AUC score.

40. How did you determine which 500 features were the most important to feed into your models?

Answer: Features were ranked by the average decrease in Gini impurity, which was used as the main splitting criterion across all 1,000 decision trees during Random Forest training .

Section 6: Results, Evaluation & Ablation

41. What AUC did the Random Forest model achieve during strict cross-platform testing?

Answer: It achieved a testing AUC of 0.669.

42. The Random Forest scored a 0.96 AUC in training but dropped significantly during testing. What does this indicate?

Answer: It indicates the highly complex ensemble slightly overfit to the platform-specific technical noise of the RNA-seq training data, failing to generalize perfectly.

43. How did the margin optimization of the SVM compare to the Random Forest's performance?

Answer: The SVM demonstrated improved generalizability over the Random Forest, achieving a higher testing AUC of 0.682, indicating it is more resilient to technical variances.

44. Which model ultimately performed the best, and what was its cross-platform AUC?

Answer: Logistic Regression (LR) performed the best, achieving a peak AUC of 0.712.

45. Why do you believe a simpler linear model (LR) outperformed a complex decision tree ensemble (RF) on this specific dataset?

Answer: Because in high-dimensional spaces with inherently subtle systemic signals, simpler weighted linear models avoid memorizing platform-specific artifacts and provide vastly superior generalization.

46. What was the exact goal of conducting the ablation study on your top-performing model?

Answer: To isolate the impact of feature count on cross-platform predictive performance and determine the optimal number of features for the ALS molecular fingerprint.

47. Based on the ablation study, what is the absolute optimal number of features required for the highest predictive efficiency?

Answer: The absolute optimal number is exactly 25 features, which yielded the peak diagnostic performance with an AUC of 0.746.

48. What happens to the model's performance when you increase the feature count to 50 or 100, and why does this occur?

Answer: The model's performance plateaus and then slightly decreases (plummeting to 0.741) because larger feature sets begin to introduce platform-specific technical noise that outweighs the added biological signal.

Section 7: Limitations & Future Directions

49. Despite your rigorous use of ComBat, what residual technical issue persisted when merging your datasets?

Answer: Residual cross-platform noise inevitably persisted when merging sequencing read counts with microarray fluorescence intensities.

50. What specific machine learning or deep learning architectures do you suggest exploring in future work to better capture non-linear multi-omics interactions?

Answer: Advanced artificial neural networks, such as autoencoders or graph neural networks.